

2025 CHEME EXPO JUDGING SHEET

Judge Name: Maxum Caesar

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

Comments:

Need to give more data about using cheaper feedstock and how much that would reduce price and increase CapEx

x

43

2025 CHEME EXPO JUDGING SHEET

Judge Name: Maxum Caesar

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	2
Total	15

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

- Would be useful for each judge to have a total PFD.
- Explain how to cycle and remove dead bacteria

35

2025 CHEME EXPO JUDGING SHEET

Judge Name: Maxum Caesar

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

50

Comments:

- Very good presentation
- Could use more heat integration - reactor shell needs boiler feed water to prevent hotspots. This can generate steam, which can be used in a turbine to generate electricity and use LPS in heat exchangers.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Maxum Caesar

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	2
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	16

33

Comments:

- * Utilities shouldn't be on a PFD.
- * Consider more heat integration
- * Flowscheme is complex, pamphlet to show sections so it's easier to follow?

2025 CHEME EXPO JUDGING SHEET

Judge Name: Ken Bicknell

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

42

Comments:

- present w/ more eye contact. Faced Board when talking about Process.
- good layout; a marketing for idea. had logo :). had PFP handout.
- economic discussion very well done. Was able to defend tough questions
- two people canceled presentation. would like to hear from others

2025 CHEME EXPO JUDGING SHEET

Judge Name: Ken Bicknell

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

Comments:

40

2025 CHEME EXPO JUDGING SHEET

Judge Name: Ken Bicknell

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	2
Total	15

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	16

31

Comments:

- doesn't seem that all ~~participated~~ participated equally. a couple carried the team.
- two of team know the material material.
- could have done better presenting.
- there is a garbage processing facility in Chicago doing the same things but it is a dirty methane stream. suggested to team to reach out to company. Southside

2025 CHEME EXPO JUDGING SHEET

Judge Name: Ken Bicknell

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

46

Comments:

Group had good dynamics. Seemed each had a part in design. Well done!

2025 CHEME EXPO JUDGING SHEET

Judge Name: Megan Young

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	22

Comments:

They had really good presentation skills.
Project was ~~very~~ very thoughtful & researched.

43

2025 CHEME EXPO JUDGING SHEET

Judge Name: Megan Young

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	17

Comments:

34

2025 CHEME EXPO JUDGING SHEET

Judge Name: Megan Young

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

Poster is very professional and informative.
Handouts were helpful to understand the process.

45

2025 CHEME EXPO JUDGING SHEET

Judge Name: Megan Young

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	19

Comments:

36

2025 CHEME EXPO JUDGING SHEET

Judge Name: Victor Nieto

Team Number: 9

Team Members: Ajit Mishra, Emily Straszewski, Elise Wojack

Technical Quality Natalie Salas

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

47

ASPer / no
PFD controls

+1 SOW
Angh
Presentation

Comments:

PFD was ASPer simulation, controls were separated from poster. Project did not have ROI after 20 years

2025 CHEME EXPO JUDGING SHEET

Judge Name: Victor Nieto

Team Number: 8

Team Members: Aluba Awan, Cassidy Bohyer, Kaleb Folan, Jenna Soshi

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

49

Comments:

Flow control input was very confusing. Had check valve on outlet of one pump and not another.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Victor Nieto

Team Number: 7

Team Members: ~~Toto Huyer~~ Catalina Toro, Huyer nguyen, Varun Maheshwari

Technical Quality Xiaofu Lin

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

49

Comments: Did not utilize heat exchanger network / use of it was unclear. very well made presentation and web page

2025 CHEME EXPO JUDGING SHEET

Judge Name: Victor Nieto

Team Number: 1

Team Members: Monica Huizar, Jonah Perlin, Jesus Raa-Valgas,

Technical Quality Carol Teran

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

49

Comments:

Very detailed. Was only group I saw utilizing heat exchanger network. A little hard to see all the info on the presentation. Used both kg/hr and lb/hr for flow rates.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 9 → Creating Renewable Natural Gas from Municipal Wastewater

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	24

Comments:

47

2025 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni →

Team Number: 7 → Emissions to Fuel

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

47

2025 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni → Fork 2 Fuel Green Solution
Team Number: 6
Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	16

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	17

Comments:

33

2025 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 5 → *Electrifying Ammonia*

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

Comments:

43

2025 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

41

2025 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

Comments:

46

2025 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

39

2025 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

40

2025 CHEME EXPO JUDGING SHEET

Judge Name: Wayshalee Patel

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	3
Total	19

Comments:

39

2025 CHEME EXPO JUDGING SHEET

Judge Name: Wayshalee Patel

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5 2
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	15

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	16

Comments:

31

2025 CHEME EXPO JUDGING SHEET

Judge Name: Wayshalee Patel

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	2
Total	14

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	19

Comments:

33

2025 CHEME EXPO JUDGING SHEET

Judge Name: Wayshalee Patel

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	16

Comments:

33

Lignocellulosic Bioethanol

2025 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh *bie*

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4.5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	23.5

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4.5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4.5
Total	24

47.5

Comments:

Great team work. All members were active in the discussion and seemed to have a good overall view of the project. I thought it was the best presentation based on the ones I judged.

BioNtro: Bacteria Driven Green Ammonia

2025 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	2
5. Project practicality in view of contemporary engineering standards/practice	3
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3.5
2. Knowledge of the background material	4.0
3. Organization & quality of presentation material	4.0
4. Group dynamics/member participation	3.5
5. Quality, effectiveness, & use of visual aids	4.0
Total	19

38

Comments:

Overall the group had a great understanding of reaction. Matthew was really engaging and passionate about it. They had some flaws in their $\text{NH}_3/\text{H}_2/\text{H}_2\text{O}$ compression system where the temperature was -35°C will water present. They acknowledged that there was an issue w/ the design. I thought their poster layout was the best.

Catalytic Pyrolysis of PE

2025 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4.5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3.5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3.5
Total	19.5

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3.5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3.5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	20

39.5

Comments:

Liked the process flow description. They put ~~into~~ a lot of effort into it. Roberto & Matthew were the strongest members. They contributed a lot and knew how to answer my questions. Their PFD was ~~almost~~ too good a little too much detail. I was hard to see ~~any~~ all of the equipment.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 3

Team Members:

Renewable Diesel Production from Waste cooking oil
& beer oil

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3.5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4.5
Total	19 20

39

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	5
Total	19

~~5.5~~
the talking

Comments:

Only project where I did not receive a brochure.
The women presenters carried the group. They pretty
much led the discussion and answered my questions.
Overall I thought it was a ~~decent~~ decent presentation.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Iris Alvarado Santiago

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	22

43/50

Comments:

Kudos for explaining that the pipeline from point A to B was excluded from the economic analysis. Transitions were good in between presenters. They were realistic on methanol pricing and if the project would be a go or no go.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Iris Alvarado Santiago

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

45/50

Comments:

They included recommendations in their poster. I would have liked to see optimization as part of the poster, but due to timing was not included; however, they were honest on that respect.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Iris Alvarado Santiago

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4.5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	22.5/25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25/25

48/50

Comments:

Should consider transportation pipeline from point A to plant or ~~mention~~ mention in the presentation the reason of why its not included. From all the teams I judge today, the dynamic and transitions were good/best

2025 CHEME EXPO JUDGING SHEET

Judge Name: Iris Alvarado Santiago

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	20/25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4 5 4 3
3. Organization & quality of presentation material	4 4 3
4. Group dynamics/member participation	4 4 3
5. Quality, effectiveness, & use of visual aids	5
Total	20/25

Only 2 members seems to know about the project

41/50

Comments:

- Economic analysis could had been improved a little.
- Only 2 students seem to know the ins/outs of the project.
- Transitions between the presenters could be improved; however this could be that they were close to a noisy room and they were presenting first.

Catalytic Production of Polyethylene

2025 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Zalc

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3.5
4. Overall fulfillment of project objectives	4.5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	16

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3.5
5. Quality, effectiveness, & use of visual aids	4
Total	19.5

Comments:

25.5

Pipe-to-pump

2025 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Zalc

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4.5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4.5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4.5
Total	23.5

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4.5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4.5
4. Group dynamics/member participation	4.5
5. Quality, effectiveness, & use of visual aids	5
Total	23.5

Comments:

47

Bio Nitro

2025 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Zalc

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4.5
4. Overall fulfillment of project objectives	4.75
5. Project practicality in view of contemporary engineering standards/practice	4.5
Total	22.75

18
4.75

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4.25
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4.5
4. Group dynamics/member participation	4.5
5. Quality, effectiveness, & use of visual aids	5
Total	23.25

19
4.25

Comments:

46

Renewable Diesel

2025 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Zalc

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4.5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4.5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4.75
2. Knowledge of the background material	4.75
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24.5

Comments:

48.5

2025 CHEME EXPO JUDGING SHEET

Judge Name: Peter Herena

Team Number: 1

Team Members:

Lignocellulose Bioethanol

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	18 20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	3
Total	19

Comments:

39

2025 CHEME EXPO JUDGING SHEET

Judge Name: Peter Herena

Team Number: 4

Team Members:

catalytic pyrolysis of Polyethylene

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4 4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3 4
Total	20 19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20 19

38

Comments:

2025 CHEME EXPO JUDGING SHEET

Judge Name: Peter Herena

Team Number: 3

Team Members:

Renewable Diesel from waste oil

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	3
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	23

43

Comments:

2025 CHEME EXPO JUDGING SHEET

Judge Name: Peter Herena

Team Number: 2

Team Members:

Bacteria-driven Green Ammonia

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

40

Comments:

2025 CHEME EXPO JUDGING SHEET

Judge Name: Suzane Carneiro

Team Number: 9

Team Members: Ajit M., Natalie S., Emily S., Elise W.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	19

36

Comments:

- Good market strategy
- Solution approach with process simplification
- Safety solutions limited to PPE and monitors
- limited/no process controls presented
- ~~limited~~ limited market assessment

2025 CHEME EXPO JUDGING SHEET

Judge Name: Suzane Carneiro

Team Number: 5

Team Members: RIA P., ERNIE R., MARILENA R., DALLAS S.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

44

Comments: - Included market assessment in the presentation
- Good ~~recommendations~~ / future work
- limited safety assessment
- good group dynamic
- limited process controls

2025 CHEME EXPO JUDGING SHEET

Judge Name: Suzane Carneiro

Team Number: 6

Team Members: RYAN B., JENNIFER L., FIONA L., JOSH O., TYLER R.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

45

Comments: - Clear problem statement

- Good / excellent presentation flow and dynamic
- Lack of proposed solutions / recommendations for the environmental concerns ~~was~~ identified
- ~~limited~~ ^{limited} market assessment
- Safety solutions limited to PPE and monitors
- limited process controls

2025 CHEME EXPO JUDGING SHEET

Judge Name: Suzane Carneiro

Team Number: 4

Team Members: Anney, Hahn T., Matthew M., Roberto A.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

47

Comments:

- Very good / detailed process diagram
- Included a hazard analysis to the project
- Included safety ^{and process} controls in the presentation
- limited market assessment
- Opportunity to improve environmental solutions

2025 CHEME EXPO JUDGING SHEET

Judge Name: Alex Chen

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

45

Comments:

overall well done, some typos, etc. on poster. recommend to focus more on key takeaways for the audience. safety could have been addressed more thoroughly on the ~~overall~~ overall process. technically sound + realistic economics!

2025 CHEME EXPO JUDGING SHEET

Judge Name: Alex Chen

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

48

Comments:

beautifully done poster in terms of formatting, digestibility/ readability. easy to follow. some of the costs re: operability + maint were not thoroughly considered.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Alex Chen

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

great definition, specific and detailed

spare reactors

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	24

no safety mention of HCl

49

Comments:

well thought out, presentation/speakers had a good flow.
liked that the objective was very realistic and that the team approached ops/maintenance ~~realistically~~ ~~practically~~ practically.

2025 CHEME EXPO JUDGING SHEET

Judge Name: Alex Chen

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	20

Solutions are a bit vague
problems are high level

slightly unrealistic economics

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

inconsistent figure labels

46

Comments:

good work overall, but could improve upon tailoring the presentation to the audience (e.g. focusing on the key takeaways, increasing readability, better use of visual aids).

↳ rev vs. cost pie chart was a format that didn't allow for easy digestibility)

also generally better to represent a more practical / realistic cost (e.g. \$250MM ^{full} grant is not really feasible)

150

2025 CHEME EXPO JUDGING SHEETName: *Emission to Fuel***Technical Quality**

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	19

Comments:

DEO

2025 CHEME EXPO JUDGING SHEET

Name: MSWS-RNG

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	20

Comments:

Name: Dennis O'Brien

2025 CHEME EXPO JUDGING SHEET

FORK TO FUEL

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

PHS

2025 CHEME EXPO JUDGING SHEET

Name: EMISSIONS TO FUEL TEAM 47

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	24

Comments:

THOROUGH DESIGN WORK

2025 CHEME EXPO JUDGING SHEET

Name: SOY TO BIO DIESEL #8

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

Comments:

GOOD WORK.

PMS

2025 CHEME EXPO JUDGING SHEET

Name: PINE-TO-PUMP LIGNOCELLULOSIC ETOL

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

Comments:

LOTS OF GOOD INFORMATION; BUT POSTER IS A LITTLE CROWDED
GOOD JOB.

P48

2025 CHEME EXPO JUDGING SHEET

Name:

ELECTRIFYING D/H TEAM 5

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

Comments:

2025 CHEME EXPO JUDGING SHEET

Name: *Fork to Fuel*

[Signature]
Danny

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4 3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	

*H₂S identified, but not handled
methane is produced but pipeline or storage not provided
wastewater not handled, sludge not handled*

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	

*Attractive poster
good colors items
sized well, easy to read*

Comments:

*Does Digester need to be agitated?
Sludge dewatering & composting missing, water recycled or disposed?
Solvent not stated
Feed waste disposal cost is "tipping fee" to land-fill EPA has typical cost.
Quantify H₂S → if may require capture maybe feed can be sorted
methane needs a compressor
Benzene needs further investigation*

Name:

Bio Nitro

2025 CHEME EXPO JUDGING SHEET

Name

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	

The process uses an expensive feedstock to produce a commodity chemical. Could a better feedstock be identified? Maybe a N-fixing bacteria & Air?

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	

Good Chemistry and Rx modeling some areas of weakness

Comments:

Pilot Plants are typ. as small as possible: (to minimize cost)

- produce enough for required analyses
- produce product/application samples
- large enough to measure all variables accurately

2025 CHEME EXPO JUDGING SHEET

Name:

Electrifying Ammanica

James

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	2
Total	

*Investigate flow schemes of existing plants in Europe & literature for feasibility
missing information on some process steps - electrolysis & cold box*

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	

*Some tables need formatting and text scaling
plots need larger text*

Comments:

*Electrical Safety missing
Recommendations should address weaknesses in process or economics or improve yields or reduce cost
payback should be "none", not 0 years
Water treatment missing from PEM feed.
PFD should show flow in/out top/bottom of Rx's
" " " all flows - O₂ & water missing
" " " Prier needed to prevent freezing in cryo section
O₂ as byproduct has value*

2025 CHEME EXPO JUDGING SHEET

Name:

Say to the World

Rance

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	

Biodiesel plants have been shutting down because Renewable diesel is better. Foodstock is limiting

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4 5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	

Very attractive layout
sizing & coloring
Flat text size could be larger

Comments: Throughput & main product flows should be mentioned
Feed pretreatment should be included
Alternative feeds can be more profitable
Recycle streams often require drag streams to remove impurities
~~A chemistry section could be helpful~~



2025 CHEME EXPO JUDGING SHEET

Name: *Pine to Pump*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

46/50

Comments:

Have a little more confidence. It's ok to say you would need to confirm, or this was based on literature sources.

Great job overall!

2025 CHEME EXPO JUDGING SHEET

Name: *Electrifying Ammonia*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4 4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	20 21

44/50

Comments:

"air energy" isn't a term of art

Make sure to proofread before printing

Good job answering questions

2025 CHEME EXPO JUDGING SHEET

Name: Emissions to Fuel: ^{Model} All via CC

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4 5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	2
5. Quality, effectiveness, & use of visual aids	5
Total	21

43/50

Comments:

Need full group to answer questions
Well thought-out concept
Impractical vessel sizes

2025 CHEME EXPO JUDGING SHEET

Name: *Soy to the World Biodiesel From Soybean Oil*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	<i>4</i>
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	<i>3</i>
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	<i>3</i>
4. Overall fulfillment of project objectives	<i>4</i>
5. Project practicality in view of contemporary engineering standards/practice	<i>4</i>
Total	<i>18</i>

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	<i>4</i>
2. Knowledge of the background material	<i>4</i>
3. Organization & quality of presentation material	<i>5</i>
4. Group dynamics/member participation	<i>5</i>
5. Quality, effectiveness, & use of visual aids	<i>5</i>
Total	<i>23</i>

need to slow down

Comments:

Liked how they talked about topics not on the poster, didn't just read the poster. Poster easy to read.

2025 CHEME EXPO JUDGING SHEET

Name: *Fats Forward: Fueling the Renewable Diesel Future*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	✓ 4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

2025 CHEME EXPO JUDGING SHEET

Name: *MSWS - RNC*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	<i>4</i>
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	<i>3</i>
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	<i>3</i>
4. Overall fulfillment of project objectives	<i>4</i>
5. Project practicality in view of contemporary engineering standards/practice	<i>3</i>
Total	<i>17</i>

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	<i>3</i>
2. Knowledge of the background material	<i>3</i>
3. Organization & quality of presentation material	<i>4</i>
4. Group dynamics/member participation	<i>4</i>
5. Quality, effectiveness, & use of visual aids	<i>4</i>
Total	<i>18</i>

Comments:

Professionalism really lacking during presentation, although it was the last time slot - getting a little slap happy?

2025 CHEME EXPO JUDGING SHEET

Name: *Pine to Pump ~~Distilled~~ Bioethanol*

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	<i>4</i>
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	<i>3</i>
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	<i>3</i>
4. Overall fulfillment of project objectives	<i>4</i>
5. Project practicality in view of contemporary engineering standards/practice	<i>3⁺</i>
Total	<i>17.25</i>

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	<i>5</i>
2. Knowledge of the background material	<i>5</i>
3. Organization & quality of presentation material	<i>5</i>
4. Group dynamics/member participation	<i>5</i>
5. Quality, effectiveness, & use of visual aids	<i>5</i>
Total	<i>25</i>

Comments:

Really liked the presentation.